Lecture 5

Imperfect Information

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POLS W4732

Imperfect Information

So far we assumed actions are observable by all players

Often, this is not realistic

Examples:

- Voters can only imperfectly see politicians' actions
  - corruption, time allocation, influence on bills
- Politicians cannot always monitor voting decisions by voters and other politicians
- Enemies/competitors can only imperfectly observe each other's military plans/investment decisions

Extensive Form Game with Imperfect Information

- Players
- The sequence of play, specifying
  - actions available for each player at each history
  - Terminal histories
- Payoffs
- **A specification of each player's information about the history at every point at which she moves**

Information Sets

Denote by H_i the set of histories after which player i moves

Specify i's information by partitioning H_i into collection of **information sets**

Player i knows which information set she is in, but not which history has occurred

Denote set of actions available after history h by $A_i(h)$

Two histories, h and h', can only be in the same information set if $A_i(h) = A_i(h')$

Basically, I am not fully sure about which node of the extensive form I am finding myself when called upon choosing an action

Extensive Form Games with Imperfect Information

Definition

An **extensive form game with imperfect information** consists of

- A set of players $N := \{1, ..., n\}$, possibly including Nature
- A set of histories $\mathcal{H} = \{h_1, ..., h_H\}$ with $\mathcal{O} \subset \mathcal{H}$ the set of outcomes
- For each player,
  - An information partition $\mathcal{I}_i$
  - Action sets $A_i(I)$ for each $I \in \mathcal{I}_i$
  - A prob. distribution over actions available to Nature (belief)
  - Preferences over the set of lotteries over terminal histories ($\Delta(\mathcal{O})$), usually expectation over $\pi_i : \mathcal{O} \to \mathbb{R}$

An Example: 1-card poker

Each of two players begins by putting a dollar in the pot

Player 1 dealt a card that is equally likely to be *High* or *Low*

Player 1 observes card, but Player 2 does not

Player 1 may *Check* or *Raise*

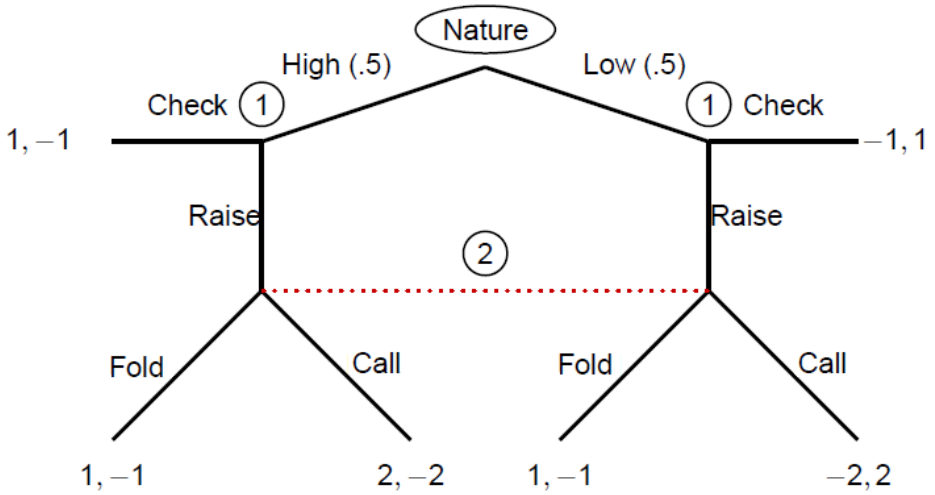
If *Check* shows card to Player 2: wins if *High*, loses if *Low*

If *Raise* adds a dollar to pot and Player 2 chooses whether to *Fold* or *Call*

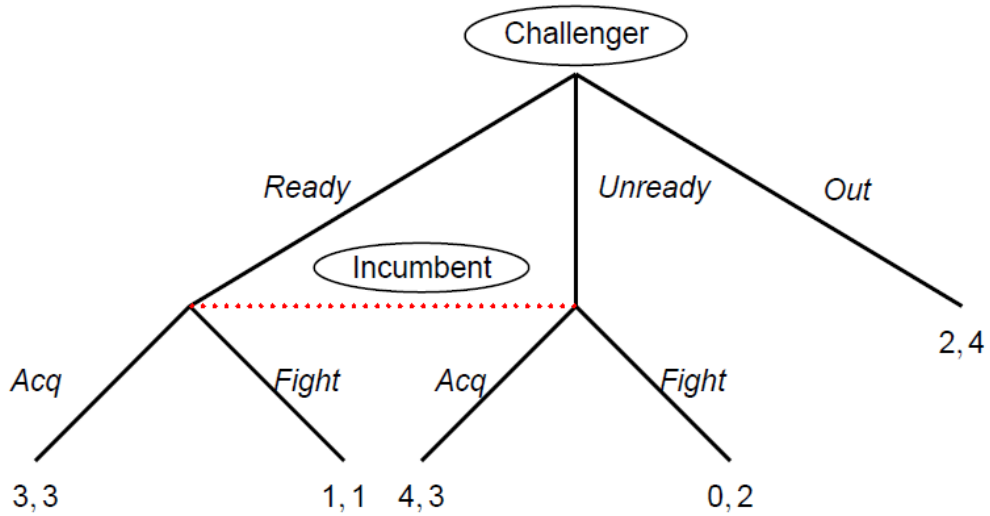
If *Fold*, Player 1 wins

If *Call* Player 2 adds a dollar and then Player shows card: 1 wins if *High* and loses if *Low*

1-card poker


Nature
High (.5)
Low (.5)
Check
1
1
Check
1, −1
−1, 1
Raise
Raise
2
Fold
Call
Fold
Call
1, −1
2, −2
1, −1
−2, 2

No move by Nature: The Entry Game with Uncertainty



Exercise Consider the perfect information version of this game:

(i) write down the associated strategic form

(ii) characterize the set of Nash Equilibria

Strategies

A strategy for a player specifies an action **at every information set** in which it is her turn to move

A complete contingent plan given available information

Definition

A *pure strategy* for player i in an extensive form game is a function that assigns to each of i's information sets I an action in $A_i(I)$

A mixed strategy is a probability distribution over pure strategies

Nash Equilibrium

Definition

A (mixed) strategy profile α^* in an extensive form game is a *mixed strategy NE* if, for each player i and every mixed strategy α_i', player i's expected payoff to α_i^* is at least as large as her expected payoff to $(\alpha_i', \alpha_{-i}^*)$

Strategic Form Representation

As for full information extensive form games, we can represent these games in strategic form and then solve for the NE

		Incumbent	
		Acq	$Fight$
Challenger	Ready	$3, 3$	$1, 1$
	Unready	$4, 3$	$0, 2$
	Out	$2, 4$	$2, 4$

Two pure strategy NE (Unready, Acq) and (Out, Fight)

Non-Credible Threats

(Out, Fight) is unrealistic (as in the perfect information entry game)

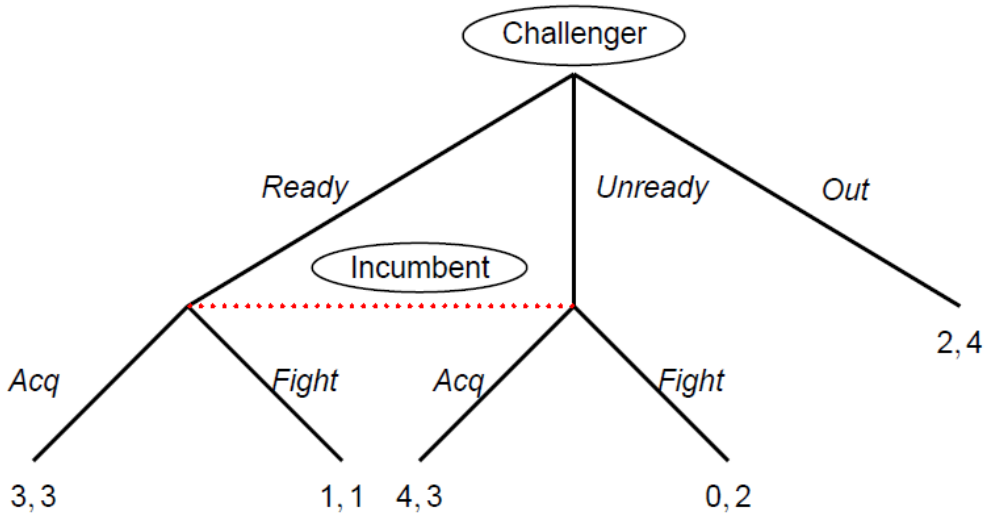
Way out under perfect information: subgame perfection

With imperfect information, natural extension is to make players optimize at every information set rather than at every subgame

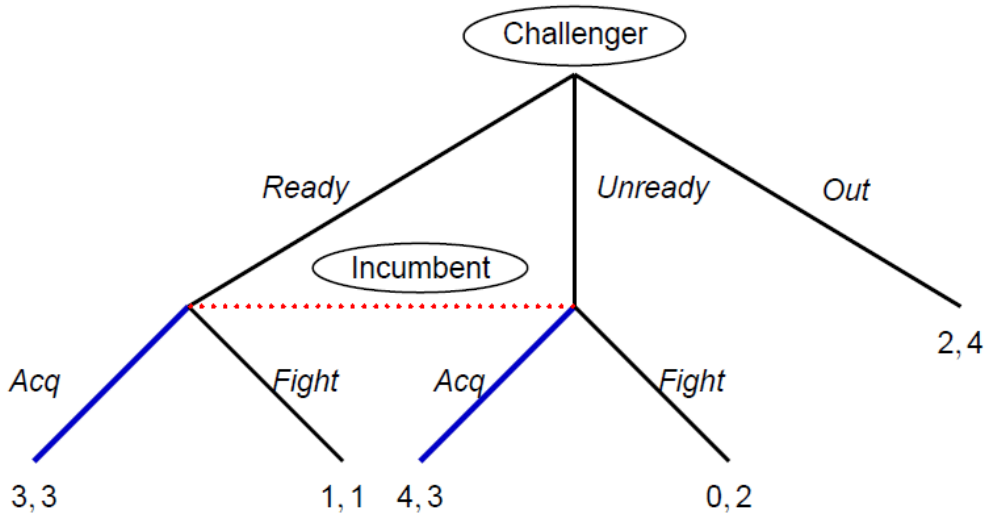
In modified entry game, clear that Fight is suboptimal at information set, **regardless of beliefs**

In this case, it is easy to 'refine out' unrealistic equilibria

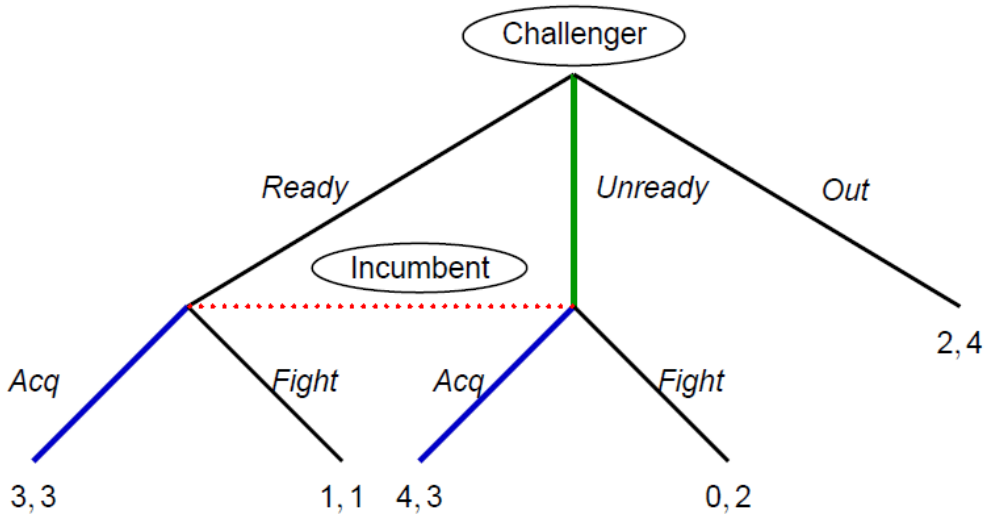
Eliminating the Non-Credible Threat



Eliminating the Non-Credible Threat

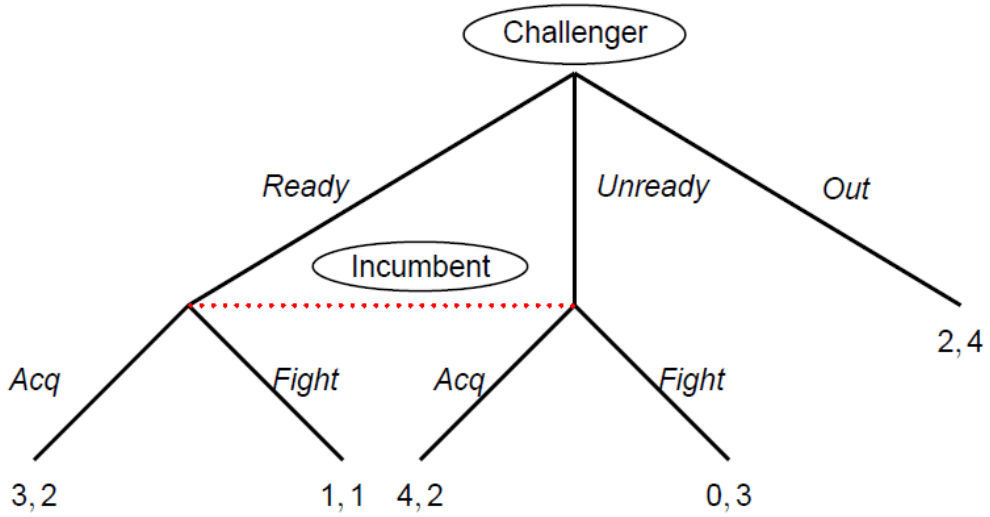


Eliminating the Non-Credible Threat



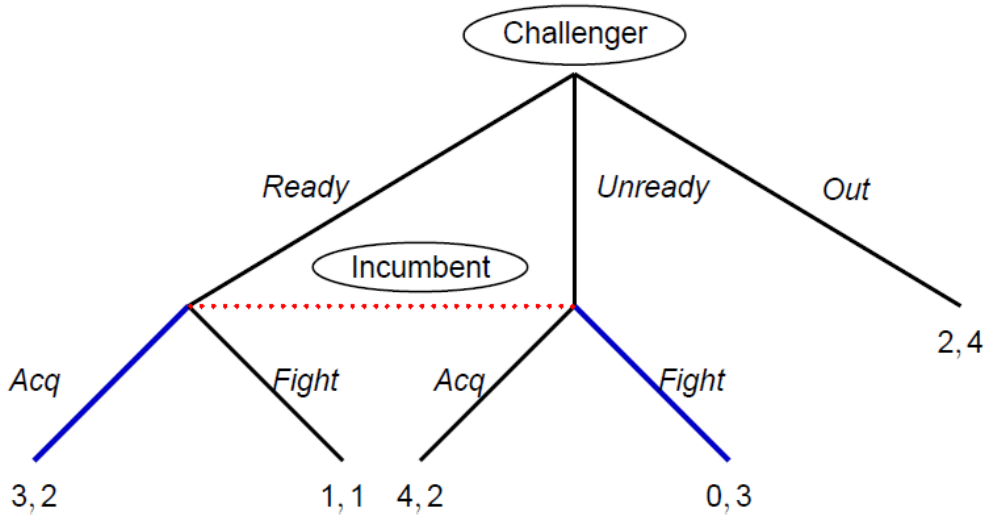
Extending Subgame Perfection is not obvious

Consider a small modification of the Entry Game



Extending Subgame Perfection is not obvious

Consider a small modification of the Entry Game



Incumbent prefers to fight against unready challenger

Optimality depends on beliefs

Beliefs and Equilibrium

NE of strategic game characterized by two requirements:

1. Each player optimizes given beliefs about other players
2. Each player's belief are correct

We want to extend this idea to extensive form games

Need to define eqm in terms of pairs of **strategy profiles and beliefs**

Beliefs

At information set containing more than one history, player whose turn it is to move forms belief about history that occurred

Model beliefs as probability distribution over histories in the information set

A collection of beliefs, one for each information set of every player, is a **belief system**

Definition

A **belief system** in an extensive form game is a function that assigns to each information set a probability distribution over the histories in that information set

Behavioral Strategies

A mixed strategy is a probability distribution over pure strategies (complete contingent plans)

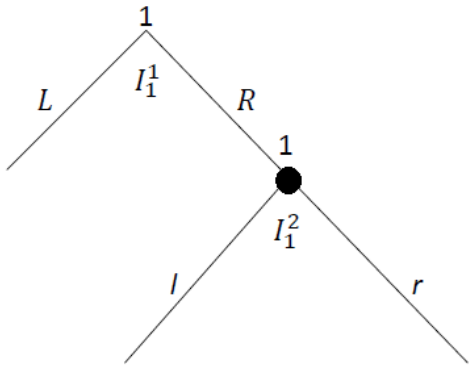
More convenient to allow players to choose probability distributions over actions available *at each information set*

Definition

A **behavioral strategy** of player i is a function that assigns to each of i's information sets I a probability distribution over the actions in $A_i(I)$, with the property that each probability distribution is independent of every other distribution

In games we will study, behavioral strategies and mixed strategies are equivalent

Behavioral and mixed strategies: An Example



- Behavioral strategy for 1 is two probability distributions, one over $\{\mathrm{L}, \mathrm{R}\}$, one over $\{\mathrm{l}, \mathrm{r}\}$
- Mixed strategy is a probability distribution over $\{(\mathrm{L},\mathrm{l}), (\mathrm{L}, \mathrm{r}), (\mathrm{R},\mathrm{l}), (\mathrm{R}, \mathrm{r})\}$

Exercise

Consider the following game, where I_2^1, I_2^2, I_2^3 denote the information sets for Pl. 2

Consider the pure strategies for Pl. 2: $s_2 \equiv (L, l, r)$, $s_2' \equiv (R, r, \ell)$, and $s_2'' \equiv (L, r, r)$

Find the behavioral strategy equivalent to the mixed strategy that puts probability $1/4$ on s_2 and $1/2$ on s_2' and $1/4$ on s_2''

Assessments and Consistency

Want to think about profiles of beliefs and strategies such that all players are optimizing at all information sets

This will extend the idea of subgame perfection to optimizing at all information sets, given beliefs

Definition

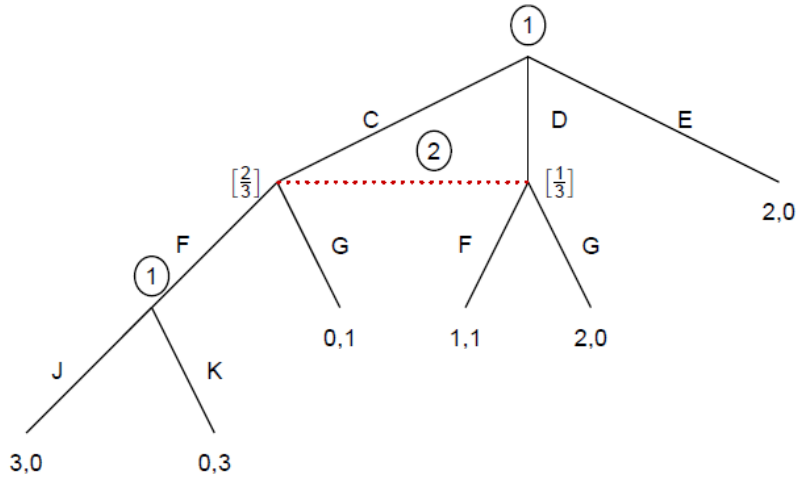
An **assessment** is a pair consisting of a profile of behavioral strategies (β) and a belief system (μ)

β is **sequentially rational** if each player's strategy is optimal whenever she has to move, **given her beliefs and the other players' strategies**

μ satisfies **consistency** with β if each player's beliefs are derived from Bayes' rule whenever possible[1]

1. For precise definition, see Fudemberg & Tirole 8.2.3

Sequential Rationality: An Example



Suppose Pl. 1 plays (E, J) and Pl. 2's beliefs as given

At Pl. 2's information set expected payoff of F is $\frac{1}{3}$ and expected payoff of G is $\frac{2}{3}$

This implies that DJ and EJ are both best responses for Pl. 1 given Pl. 2's beliefs and strategy

Sequential Rationality

Let (β, μ) be an assessment in an arbitrary game

Let I be an information set of player i

Denote by $O_I(\beta, \mu)$ the probability distribution over terminal histories resulting from

- each history in I occurring with the probability assigned to it by i's belief μ_i
- players adhering to the behavioral strategy profile β

Sequential rationality requires that for each player i *and each of her information sets* I,

$$U_i[O_I(\beta, \mu)] \geq U_i[O_I((\alpha_i, \beta_{-i}), \mu)]$$

Consistency of Beliefs with Strategies

Probability assessment assigns to any history must be probability history occurs under assessment

At an information set reached with positive probability under the assessment, beliefs determined by Bayes' Rule

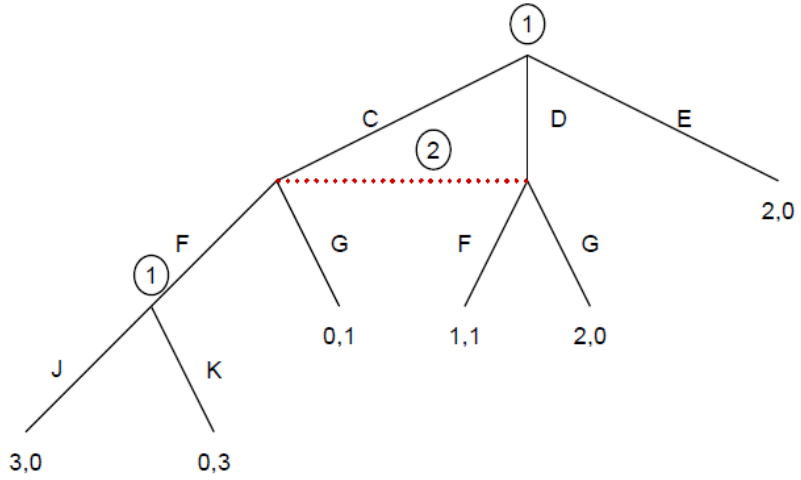
What about information sets not reached with positive probability?

- Every history in such an information set has probability 0
- **Allow player to hold any belief at such information sets**

Consistency only restricts belief system at information sets reached with positive probability. For any such history, h^*

$$\frac{\Pr(h^* \text{ according to } \beta)}{\sum_{h \in I} \Pr(h \text{ according to } \beta)}$$

Consistency: An Example



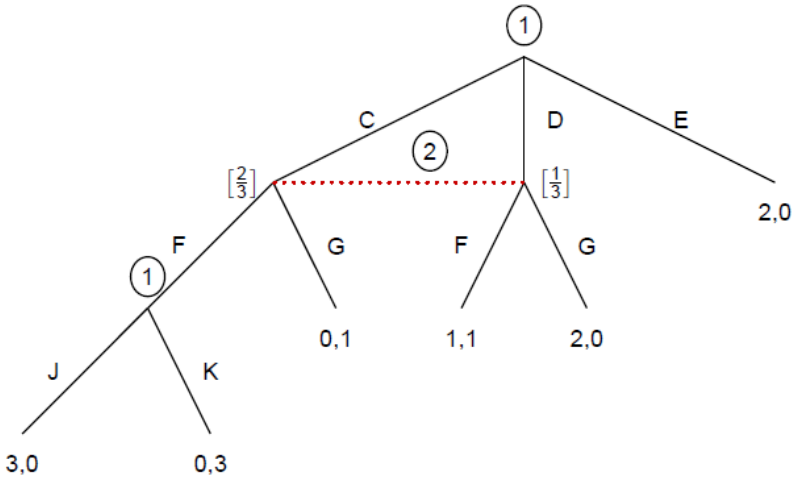
- If 1's behavioral strategy assigns probability 1 to E, then consistency places no restriction on 2's beliefs
- If 1's strategy assigns probability p to C and q to D, then 2's beliefs (at her information set) assign probability $\frac{p}{p+q}$ to C and $\frac{q}{p+q}$ to D.

Perfect Bayesian Equilibrium

Definition

An assessment (β, μ), consisting of a behavioral strategy profile β and a belief system μ, is a **(weak) Perfect Bayesian Equilibrium (PBE)** if it satisfies Sequential Rationality and Consistency of Beliefs with Strategies

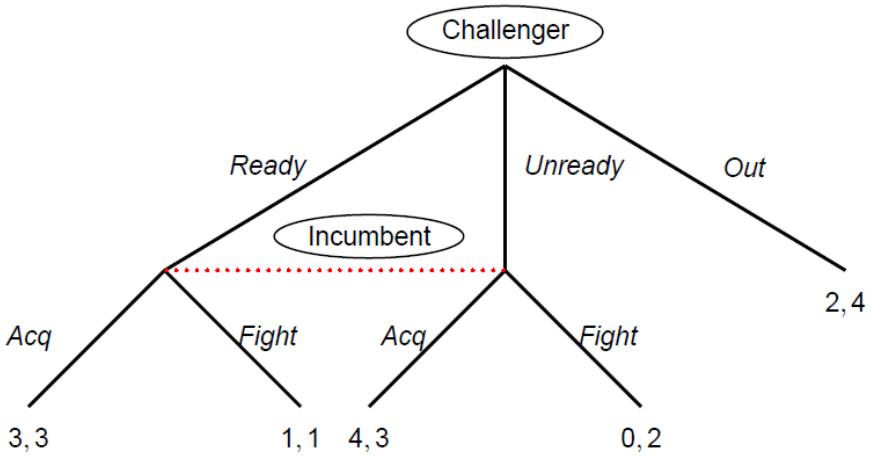
Finding a PBE



Exercise

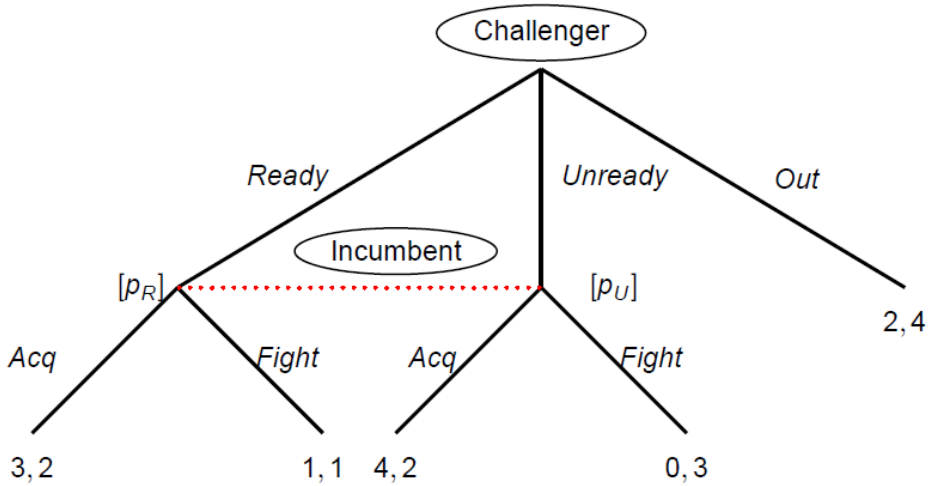
- Find the set ot sequentially rational profiles for this belief system
- Can they all be part of a PBE?

Does PBE Rule Out Non-Credible Threats?



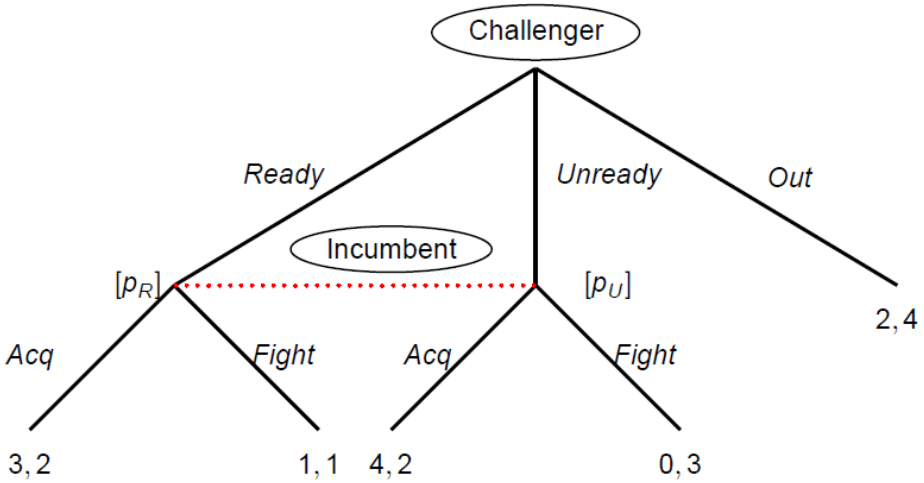
- Two NE: (U, A) and (O, F).
- Under (U, A) consistency requires Incumbent to believe U for sure, so A is optimal
- Under (O, F) consistency does not pin down beliefs, but regardless of beliefs, F is not optimal, so sequential rationality

The Modified Entry Game



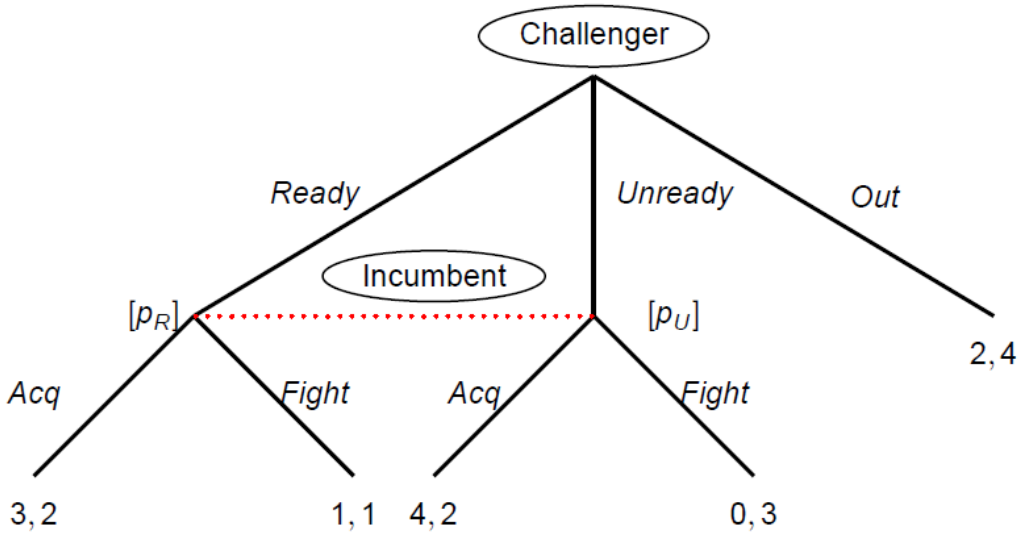
- Can there be a PBE with O for certain?
- O is optimal if expect F
- F is optimal if $p_U \times 3 + p_R \times 1 \geq 2$ or $p_U \geq \frac{1}{2}$
- Since beliefs are unrestricted, there are PBE $(O, F, p_U \geq \frac{1}{2})$

The Modified Entry Game



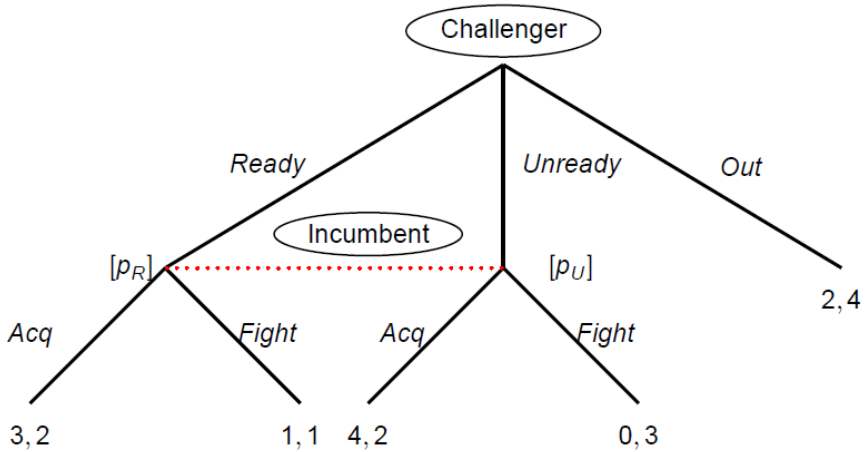
- Can there be PBE with positive probability of U but not R?
- U is optimal if A
- A is optimal if $p_U \leq \frac{1}{2}$
- But consistency requires $p_U = 1$, so no PBE

The Modified Entry Game



- **Exercise** Can there be PBE with positive probability of R but not U?

The Modified Entry Game



- Incumbent willing to mix if $p_U = \frac{1}{2}$ (let q_A be prob. of A)
- Challenger indifferent between R and U if $3q_A + 1 - q_A = 4q_A$ or $q_A = \frac{1}{2}$ (let q_R be prob. of R)
- If $q_A = \frac{1}{2}$, exp. payoff from R or U equals payoff from O

Why Refine PBE?

Bayes' rule pins down beliefs only on the equilibrium path

- Off-path beliefs may support a PBE by assigning probability to types with no reason to deviate
- The Intuitive Criterion removes types that could never gain from the deviation
- D1 favors types for which the deviation is profitable under a wider set of receiver responses

Intuitive Criterion: Beer-Quiche Game

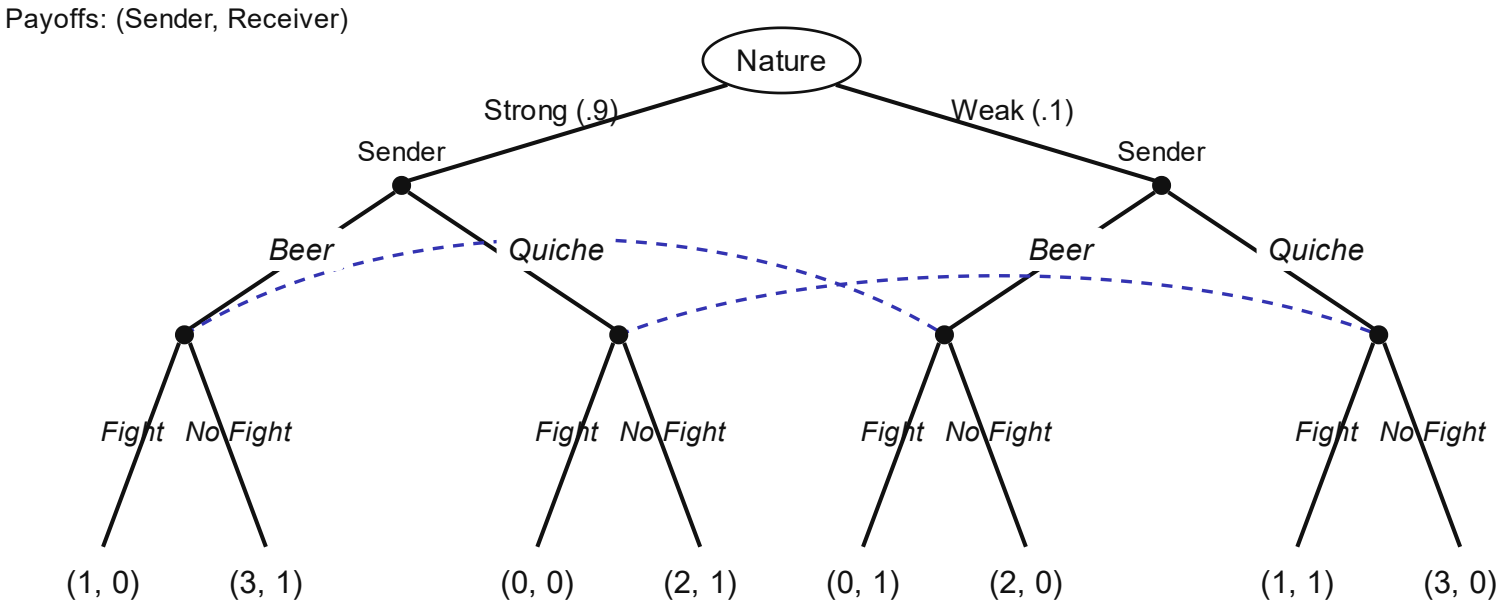
Sender type: Strong ($.9$) or Weak ($.1$)

- Strong prefers Beer; Weak prefers Quiche. The preferred breakfast adds 1 to the Sender's payoff
- No Fight gives the Sender 2; Fight gives the Sender 0, before the breakfast bonus
- The Receiver prefers Fight against Weak and No Fight against Strong

Consider the pooling PBE in which both types choose Quiche

- The Receiver chooses No Fight after Quiche and Fight after the off-path Beer
- Equilibrium payoffs are 2 for Strong and 3 for Weak

Beer-Quiche Game: Extensive Form



Blue dashed arcs connect the Receiver's information sets: breakfast observed, type unobserved

Intuitive Criterion: Off-Path Beliefs

Payoffs from the off-path Beer:

- Weak gets 2 under No Fight and 0 under Fight, both below its equilibrium payoff of 3
- Strong gets 3 under No Fight and 1 under Fight, so it gains only under No Fight

The Intuitive Criterion assigns probability zero to Weak after Beer

- The Receiver infers Strong and chooses No Fight
- Strong deviates to Beer and raises its payoff from 2 to 3

Conclusion: The pooling-on-Quiche PBE is eliminated

D1: A Three-Type Education Game

Worker type $\theta \in \{L, M, H\}$ has productivity $y_\theta \in \{1, 2, 3\}$, with equal prior probabilities

- The Worker chooses education $e \in \{0, 1\}$
- A competitive Employer observes e and pays $w = \mathbb{E}[y_\theta \mid e]$
- Worker payoff is $w - c_\theta e$, where

$$c_H = .25, \qquad c_M = .75, \qquad c_L = 2.5$$

Consider a pooling PBE with $e = 0$ and $w = 2$

- After the off-path $e = 1$, beliefs concentrated on M imply $w = 2$
- Given that wage, no type wants to deviate

D1: Profitable Response Sets

After $e = 1$, an Employer best response can produce any wage $w \in [1, 3]$

Let R_θ be the set of wages that make $e = 1$ profitable for type θ:

$$R_H = (2.25, 3], \qquad R_M = (2.75, 3], \qquad R_L = \varnothing$$

- The Intuitive Criterion removes L, but leaves both H and M as possible deviators
- D1 uses $R_M \subset R_H$ to put probability one on H after $e = 1$
- The Employer pays $w = 3$, so H earns $2.75 > 2$ and deviates

Conclusion: D1 eliminates the pooling PBE